
SC/DPS Git Management

A clear repository plan around official references, DAQ ZMQ/protobuf interfaces, OPC-UA bridges, safety checks, and release evidence

Manuel Arroyave

DUNE slow-controls software discussion – 20 August 2026



Start From Official Sources

SOURCE OF RECORD

- ▶ Requirements and interfaces come from approved EDMS and official DUNE documents.
- ▶ Git repositories implement, test, package, and document software behavior against those sources.
- ▶ Unlinked notes, prototype repos, and slides are evidence or proposals, not authority.
- ▶ Each production repo should point to the exact official source it implements.

Practical rule

A production change needs an official reference, a contract version, and evidence that the dependent clients still work.

Initial references

Use online EDMS/DUNE links only. The next slide is the starter reference register.

Reference Register We Should Maintain

ALL EXAMPLES ARE CLICKABLE ONLINE LINKS

Reference class	Examples	What the Git repos record
SC/DPS and subsystem ICDs	EDMS 3309688 (PDS SC), EDMS 3362709 (BDE), EDMS 3315615 (HVS), EDMS 3297045 (TDE), EDMS 3449868 (DAQ), EDMS 2401090 (DPS).	Which variables, commands, permits, interlocks, and failure behavior are in scope.
HWDB and installation identity	EDMS project CERN-0000279681 (HWDB exports), EDMS 3357086 (PDS to HWDB), rack/network/interface records linked from EDMS.	How software maps tags and endpoints to real installed identities without duplicating live inventory.
Official software sources	DUNE-DAQ/daphnmodules , DUNE-DAQ/daphne-firmware , DUNE-DAQ/dune-wib-firmware , canonical board-service repos once chosen.	Which code owns each producer, consumer, schema, build artifact, and release tag.
Release and evidence records	System release BOM, VST reports, compatibility matrix, signed or traceable tags.	Which versions were tested together and what rollback path is approved.

Systems We Are Considering

Device producers

DAPHNE and WIB firmware, board services, ZMQ/protobuf actors, GIZMo OPC-UA producer, support-equipment adapters.

producer owns truth

Consumers and projections

DUNE DAQ controllers/opmon, OPC-UA bridges, Ignition Gateway projects, dashboards, archival paths.

same contract

Verification and deployment

Fleet emulator, vertical-slice tests, release BOM, host services, network/port policy, rollback procedure.

tested set

Boundary statement

The repository plan follows these interfaces. It is not a proposal to merge all code into one repository or to replace existing DUNE-DAQ ownership.

Why This Organization

THE PROBLEM

- ▶ DAQ and SC both talk to board services, often through the same ZMQ/protobuf contracts.
- ▶ Bridges can accidentally carry private schema copies and drift from DAQ behavior.
- ▶ Firmware, board services, bridges, Ignition exports, and deployment do not release on the same cadence.
- ▶ Safety-related assumptions can be hidden in code unless they are tracked as explicit gates.

THE ORGANIZATION

- ▶ Put contracts and compatibility policy where everyone can review them.
- ▶ Keep device implementation beside the teams that own the device behavior.
- ▶ Put cross-system tests and release BOMs in a place whose job is integration evidence.
- ▶ Keep secrets and site-specific runtime values outside public Git.

System and Interface Map



What We Need to Track

Thing to track	Minimum record	Why it matters
Official reference	EDMS/DUNE document ID, revision, owner, interpreted scope.	Prevents local assumptions from becoming silent requirements.
Contract identity	Schema version, descriptor hash, producer repo/tag, compatible consumers.	Makes DAQ, bridge, emulator, and deployment compatibility reviewable.
Command authority	Who may write in each lifecycle state, stale limits, rejection behavior, audit fields.	Prevents DAQ/SC races and makes unsupported writes fail closed.
Safety assumptions	Permit source, interlock owner, failure survival claim, reset/rollback path.	Prevents monitoring software from being credited with independent protection.
Evidence	CI logs, emulator scenarios, one-board vertical-slice results, release BOM.	Shows the selected versions worked together before deployment.

Safety and Dependency Checks in Git

Safety checks

No secrets or site credentials. Writes gated by lifecycle and fresh permits. Unsafe command paths fail closed. DPS/protection claims require the designated owner and evidence that the action survives the initiating failure.

Dependency checks

Schema hash compatibility. Producer/consumer matrix. Emulator scenario with real DAQ ZMQ client and real OPC-UA bridge. Release BOM validated before deployment.

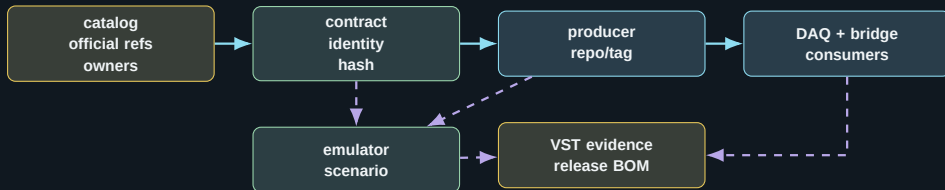
Review gate

A contract-changing pull request should link the official reference, affected producers, DAQ consumers, bridges, emulator scenarios, deployment impact, test evidence, and rollback plan.

Repository Roles

Repo role	Owns	Does not own
scdps-coordination	Reference register, matrix, ADRs, RACI, release compatibility table.	Device implementation or live inventory.
scdps-contracts	Common status semantics, schema policy, descriptor publication, compatibility tooling.	Every device register or every command.
daphne-sc-service / wib service	Board actor behavior, ZMQ/protobuf producer, firmware/service readiness.	Ignition project resources or cross-system release evidence.
daphne-opcua-bridge / wib-opcua-bridge	Device-specific OPC-UA mappings, quality/staleness, methods, bridge tests.	Private schema forks or independent protection decisions.
scdps-emulator	Production-like protocol behavior, fault scenarios, protocol-source pinning.	Production HMI or production DPS logic.
scdps-system-tests / deployment	VST orchestration, evidence reports, release BOM, host services, rollback.	Secrets, live operational databases, or HWDB duplication.

Adoption Path



Online candidates to classify

marroyav/daphne_slow_controls_plan, marroyav/daphne-sc, ecrystal/daphneZMQ, BenLand100/wib_opcua, marroyav/sc_emu, marroyav/sc_interface, marroyav/GIZMo, marroyav/gizmo-ignition.

Decisions for Thursday

- ▶ Where is the governed namespace: DUNE, DUNE-DAQ, or a coordinated cross-organization collection?
- ▶ Which repository owns the canonical DAPHNE and WIB ZMQ/protobuf contracts?
- ▶ Who are the first CODEOWNERS for references, contracts, DAQ interfaces, bridges, deployment, and evidence?
- ▶ Which vertical slice proves the first DAQ ZMQ client plus OPC-UA bridge path?
- ▶ What branch protection, tag, and release-BOM policy is the minimum before wider onboarding?

Meeting output

Approve the catalog structure and name the owners for the first two contracts.

First technical proof

Run one DAQ ZMQ client and one OPC-UA bridge against the same emulator scenario in CI.